SUVALAKSHMI S

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PROFESSIONAL SUMMARY

Dynamic and motivated M.E. graduate in Biometrics and Cyber Security with strong expertise in deep learning, image processing, and blockchain-based security systems. Proficient in Python and C, with practical knowledge of malware analysis, network security, and penetration testing tools. Demonstrated ability to design and implement innovative, research-driven solutions for real-world problems, with multiple published journal articles and conference papers to credit.

EDUCATION

M.E — Biometrics and Cyber Security 2023 – 2025
PSG College of Technology, Coimbatore | CGPA: 7.37

B.E — Electronics and Communication Engineering 2019 – 2023
National Engineering College, Kovilpatti | CGPA: 8.0

HSC — The Lakshmi Mills Higher Secondary School, Kovilpatti 2019
Score: 74.17%

SSLC — The Lakshmi Mills Higher Secondary School, Kovilpatti 2017
Score: 92.60%

TECHNICAL SKILLS

Languages	C, Python
Security	Malware Analysis, Network Security, Vulnerability Assessment & Penetration Testing
Tools	IDA, Wireshark, Autopsy, Cisco Packet Tracer
Domains	Deep Learning, Blockchain Security, Image Processing
Soft Skills	Hardworking, Punctual, Detail-oriented, Team Player

PROJECTS

Wavelet-Based Fusion for Diabetic Retinopathy Classification using Transformer Models 2025

- Designed and implemented a deep learning solution for diabetic retinopathy classification using Vision and Swin Transformer models.
- Incorporated wavelet-based image fusion to enhance retinal images by improving lesion visibility and capturing both local and global features.
- Solution supports early diagnosis and aids clinical decision-making with improved accuracy.

Fortifying Oximeter Data Protection: Safeguarding with Frodokem and NTRU Encryption 2024

- Developed a blockchain-based system to secure oximeter data using Frodokem and NTRU post-quantum encryption methods.
- Combined advanced cryptographic algorithms with blockchain immutability to ensure data integrity and protection against unauthorized access.

Underwater Image Enhancement and Classification using Deep Learning 2023

- Implemented wavelet fusion to enhance underwater image quality by merging complementary information from multiple images.
- Applied YOLOv5 for classifying enhanced images into various categories with improved detection accuracy.

PUBLICATIONS & ACHIEVEMENTS

- Published journal article: "Visual Quality Improvement in Deep Sea Images" — Journal of Pharmaceutical Negative Results, 2023.
- Research paper: "Fortifying Oximeter Data Protection: Safeguarding with Frodokem and NTRU Encryption" — Published at IEEE World Conference on Applied Intelligence and Computing, Gwalior, 2024.
- "Diabetic Retinopathy Classification using Transformer Models: A Comprehensive Survey" — Selected for International Conference on Computer Vision and Robotics, 2025.
- Participated in Codeathon – 2024.

CERTIFICATIONS

- Sololearn — C Programming Course

LANGUAGES

Tamil (Working Proficiency) • English (Working Proficiency) • Telugu (Speaking Proficiency)